



CHAPTER

3

Chemical Kinetics

Sr. No.	Concept	Formulas	Other information
1.	Instantaneous rate	Instantaneous rate $= \frac{dx}{dt} \text{ mol L}^{-1} \text{ s}^{-1}$	Here, dx is small change in concentration and dt is the smallest interval of time
2.	Average rate	Average rate $= \frac{\Delta x}{\Delta t} \text{ mol L}^{-1} \text{ s}^{-1}$	Here, Δx is change in concentration and Δt is large interval of time
3.	Rate of reaction For a reaction; $n_1 A + n_2 B \rightarrow m_1 C + m_2 D$	$\text{Rate} = - \frac{\Delta[\text{Reactants}]}{\text{time}}$ $\text{Rate} = + \frac{\Delta[\text{Products}]}{\text{time}}$ $\text{Rate} = \frac{1}{n_1} \cdot \frac{-d[A]}{dt} = \frac{1}{n_2} \cdot \frac{-d[B]}{dt}$ $= \frac{1}{m_1} \cdot \frac{+d[C]}{dt} = \frac{1}{m_2} \cdot \frac{+d[D]}{dt}$ $\text{Rate of disappearance of A} = - \frac{d[A]}{dt}$ $\text{Rate of disappearance of B} = - \frac{d[B]}{dt}$ $\text{Rate of appearance of C} = + \frac{d[C]}{dt}$ $\text{Rate of appearance of D} = + \frac{d[D]}{dt}$	Here, [A] and [B] are the molar concentration of reactants A and B respectively. [C] and [D] are the molar concentration of products C and D respectively. n_1, n_2, m_1 & m_2 are the stoichiometric coefficients of A, B, C & D in the balanced chemical equation respectively. Unit of rate of reaction is $\text{mol L}^{-1} \text{ s}^{-1}$
4.	Order of Reaction $m A + n B \rightarrow \text{products}$	$\text{Rate} = k[A]^x[B]^y$ The overall order is $x + y$.	Here, x and y are orders with respect to molar concentrations of A and B respectively. k is rate constant.

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5.	Zero Order Reaction: $R \rightarrow \text{products}$ (a) Integrated Rate Equations (b) Half-Life time ($t_{1/2}$)	$[R] = -kt + [R]_0$ $t_{1/2} = \frac{[R]_0}{2k}$	Here, 'k' is rate constant, $[R]_0$ is initial molar concentration, $[R]$ is final molar concentration after time 't' Unit of 'k' is $\text{mol L}^{-1}\text{s}^{-1}$
6.	First Order Reaction: $R \rightarrow \text{products}$ (a) Integrated Rate Equations (b) First order equation for gas phase reaction (c) Half-Life time ($t_{1/2}$)	$\ln[R] = \ln[R]_0 - kt$ $k = \frac{1}{t} \ln \frac{[R]_0}{[R]}$ $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$ $k = \frac{2.303}{t} \log \left(\frac{P_i}{2P_i - P_t} \right)$ $t_{1/2} = 0.693/k$ $t_{75\%} = 2 \times t_{1/2}$ $t_{100\%} = \infty$	Here, 'k' is rate constant, $[R]_0$ is initial molar concentration, $[R]$ is final molar concentration after time 't' $t_{75\%}$ is the time at which 75% reaction completed and $t_{100\%}$ is the time at which 100% reaction completed Unit of 'k' is s^{-1} P_i is initial pressure. P_t is pressure at time 't'.
7.	Second order reaction: $R + R \rightarrow \text{products}$ (two molecules of the same reactant) (a) Integrated Rate Equations (b) Half-Life time ($t_{1/2}$)	$\frac{1}{[R]} = kt + \frac{1}{[R]_0}$ $(t_{1/2}) = \frac{1}{k[R]_0}$	Here, $[R]_0$ is initial molar concentration of reactant. k is rate constant. $[R]$ is molar concentration of reactant at time t. k is rate constant. Unit of k: $\text{mol}^{-1} \text{L s}^{-1}$.

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8.	Arrhenius Equation: (a) For a reaction at two different temperatures (b) Temperature coefficients	$k = Ae^{\frac{-E_a}{RT}}$ $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$ $2 \text{ or } 3 \approx \frac{k_{T+10}}{k_T}$	Here, k is the rate constant. A is the pre-exponential factor or frequency factor. E _a is the activation energy. R is the universal gas constant. T is the temperature in Kelvin. k ₁ is rate constant at temperature T ₁ k ₂ is rate constant at temperature T ₂ K _T is rate constant at temperature T(°C). k _{T+10} is the rate constant at temperature T+10(°C).
9.	Enthalpy of the reaction	$\Delta H = H_p - H_R = (E_a)_f - (E_a)_b$	Here, (E _a) _f and (E _a) _b are activation energy of forward and backward reactions respectively.
10.	Unit of k	Units of $k = \frac{\text{mol L}^{-1} \text{s}^{-1}}{(\text{mol L}^{-1})^n} = \text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$	NA
11.	Collision theory of chemical reactions	$\text{Rate} = PZ_{AB} e^{\frac{-E_a}{RT}}$	Here, Z _{AB} is collision frequency of reactants A and B. E _a is the activation energy. R is the universal gas constant. T is the temperature in Kelvin. P is the steric factor